Synthetic Biology: Engineering with Living Cells

How genes, parts and circuits are designed — and how a bacterium can become a sensor

1. What is synthetic biology?

Synthetic biology applies engineering ideas — standard parts, design, testing — to living cells. Instead of moving a single gene around by trial and error, synthetic biologists try to design genetic circuits from well-understood parts, much as an electronics engineer builds a circuit from resistors and switches.

Engineers work in a cycle: Design → Build → Test → Learn, and then repeat. A circuit rarely works the first time; the data from testing show what to change.

CONNECT THE SESSIONS
Day 1: a gene is copied (transcription) and read (translation) — the circuit's basic job is to make a protein. Day 2: the lac operon is a natural genetic circuit with a sensor (the repressor) and an output (the enzymes), and quorum sensing is natural cell-to-cell communication. Day 3: plasmids and cloning are how we build the circuits.

2. The parts

A basic gene on a plasmid is made of standard parts placed in a fixed order:

Promoter	→	**Ribosome-binding site (RBS)**	→	**Coding sequence (gene)**	→	**Terminator**

Part	Job	Think of it as...
Promoter	Where RNA polymerase starts. Strong promoters give a lot of mRNA; some respond to a signal.	The on/off switch or dimmer
RBS	Where the ribosome attaches to the mRNA. Its strength controls how much protein is made from each mRNA.	The volume knob
Coding sequence	The gene itself — it codes for the protein.	The recipe
Terminator	Tells RNA polymerase to stop.	The full stop

Parts can be reporter genes that make a visible signal, such as GFP (green fluorescent protein), lacZ (a blue colour with X-gal) or coloured pigment proteins. They tell us when a circuit is active.

STANDARD PARTS AND IGEM
The iGEM competition (International Genetically Engineered Machine) began at MIT in 2003. Student teams around the world design, build and test genetic systems, and share their standard parts, called BioBricks, in an open registry so that others can reuse them. Teams are also judged on safety, how well they involve the community, and the ethics of their work.

3. Genetic circuits

- **Switches:** a promoter that responds to a chemical (like the lac promoter and IPTG) gives an ON/OFF output.
- **Toggle switch (2000):** two genes that repress each other, so the cell "remembers" one of two states — like a memory bit.
- **Repressilator (2000):** three repressors in a ring that make a protein level rise and fall in regular cycles — a bacterial clock.
- **Logic gates:** circuits that give an output only if *both* inputs are present (AND) or *either* is present (OR).
- **Quorum-sensing circuits:** engineered cells respond only when the population is dense (see Day 2).

4. Biosensors

A biosensor uses a living cell (or a biological molecule) to detect something and to report it. Every biosensor has the same structure:

INPUT **the thing to detect (e.g. arsenic in water)**	→	**SENSOR** **a promoter or protein that responds to the input**	→	**OUTPUT** **a signal we can see (colour, glow, smell)**

Example: Early iGEM teams, such as Edinburgh in 2006, built bacteria that detect arsenic in drinking water. A promoter that switches on when arsenic is present controls a reporter gene, so the bacteria change colour in contaminated water.

Advantages and limits

- Advantages: cheap, can be very sensitive, and cells copy themselves.
- Limits: slow response (hours), variability, living cells need care, and any release into the environment must be controlled.

5. Responsibility and safety

- **Containment.** Engineered cells must stay in the lab, or inside a closed device. Teams design a **kill switch** that makes the cells die outside the intended environment, and test it.
- **Safe organisms.** Choose harmless lab strains, and avoid parts from dangerous pathogens.

- **Ethics.** Ask who benefits, who could be harmed, and who has been consulted. Talk to the people affected — for example local communities, workers and experts — before and during the design.
- **Honesty.** Report failures as well as successes.

ACTIVITY: DESIGN A BIOSENSOR ON PAPER (25 MINUTES)

In teams, choose a real problem (for example: sugar-sweetened drinks, lead in water, a pollutant in a river, a toxin in food).

1. **Input:** what should be detected? Who would use the sensor?
2. **Sensor part:** what would respond to it? (Real teams search the literature for a promoter or protein.)
3. **Output:** a colour, a glow, or a smell? How will people read it?
4. **Circuit:** draw the four-part gene (promoter → RBS → reporter → terminator) and the plasmid with an origin of replication and a marker.
5. **Safety:** how will you keep the cells contained, and how will you dispose of them?
6. **People:** who should you talk to before building it?

Teams present in 2 minutes. The class asks one question about safety and one about who it helps.

Parts-card game (10 minutes)

Print the four part names on cards (Promoter, RBS, Gene, Terminator) with extras of each. Teams must place them in a working order. Then give the "mistake" cards: a gene with no promoter (nothing is made), no terminator (RNA runs on), no RBS (mRNA made but little protein). Ask students to predict what goes wrong for each.

Check your understanding

1. Name the four basic parts of a gene on a plasmid and give the job of each.
2. What is the difference between the *input* and the *output* of a biosensor?
3. Why is a kill switch used?
4. How is the lac operon like a genetic circuit?

Answers: (1) promoter — start; RBS — ribosome attachment; coding sequence — protein recipe; terminator — stop. (2) Input is the thing detected; output is the visible signal. (3) To make engineered cells die outside the intended environment, keeping them contained. (4) It senses lactose and glucose (inputs) and switches on the genes for lactose use (output).